

Experiment : 5

Hadoop Installation, Configuration, and HDFS Operations

Mapped CO(s) : CO2

Aim

To install and configure Hadoop in a single-node environment and perform basic HDFS operations using command-line tools.

Objectives

- To understand the Hadoop ecosystem and HDFS.
- To install and configure Hadoop.
- To start Hadoop services.
- To perform basic HDFS file operations.
- To verify the successful execution of HDFS commands.

Prerequisites

- Basic knowledge of Linux commands.
- Java Development Kit (JDK) installed.
- Ubuntu/Linux operating system or Virtual Machine.
- Hadoop software package.
- Terminal or Command Prompt.

Theory/Background

Apache Hadoop is an open-source framework used for storing and processing large volumes of data across distributed systems. It enables reliable, scalable, and fault-tolerant data storage and computation.

The main components of Hadoop are:

- **HDFS (Hadoop Distributed File System):** Stores data across multiple machines.
- **MapReduce:** Processes large datasets in parallel.
- **YARN (Yet Another Resource Negotiator):** Manages cluster resources and job scheduling.
- **Hadoop Common:** Provides common utilities and libraries.

HDFS stores files by dividing them into blocks and distributing them across multiple nodes. It provides high fault tolerance through data replication.

Common HDFS operations include creating directories, uploading files, listing files, viewing file contents, downloading files, copying files, and deleting files.

Algorithm/Methodology

- Install Java and Hadoop.
- Configure Hadoop environment variables.
- Configure Hadoop XML files.
- Format the Hadoop NameNode.
- Start Hadoop services.
- Verify HDFS is running.
- Create directories in HDFS.
- Upload files to HDFS.
- Display HDFS file contents.
- Download and delete files from HDFS.

Dataset Description

Dataset Used: Sample Text File

File Name: sample.txt

Description:

A simple text file is used to demonstrate HDFS operations such as uploading, listing, reading, copying, downloading, and deleting files.

Program/Implementation

Typical commands include:

- Check Hadoop version
- Format NameNode
- Start DFS services
- Create HDFS directories
- Upload files
- List files
- Read files
- Copy files
- Download files
- Delete files

Test Cases

Test Case	Expected Result
Check Hadoop version	Hadoop version displayed
Start Hadoop services	Services start successfully
Create HDFS directory	Directory created
Upload file	File uploaded to HDFS
List files	Files displayed
View file	File contents displayed
Download file	File copied to local system
Delete file	File removed from HDFS

Experimental Results

Observation

- Hadoop was successfully installed and configured.
- HDFS services started successfully.
- Directories were created in HDFS.
- Files were uploaded, viewed, and downloaded successfully.
- HDFS commands executed without errors.

Result

Hadoop was successfully configured, and all basic HDFS operations such as creating directories, uploading, listing, reading, downloading, and deleting files were performed successfully.

Conclusion

The experiment demonstrated the installation and configuration of Hadoop in a single-node environment. Basic HDFS operations were successfully performed using Hadoop shell commands, providing practical understanding of distributed storage and file management in Hadoop.

Exercises/Assignments

1. Create a new directory in HDFS and upload multiple files.
2. Copy a file from one HDFS directory to another.
3. Delete a directory from HDFS.
4. Display the contents of an HDFS file.
5. Check the available space in HDFS.